SIMULATION IN FINANCE



FRE 6103

Lecture 4 / Chapters 5-6

Professor David Shimko

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TODAY'S TOOLKIT

- Simulating any random variable (the Inverse CDF Transform Method)
- Estimating the expected value of functions of a random variable
- Efficiency in simulation (univariate)
  - Comparison to numerical integration
  - Error ranges in simulation estimates
  - Variance reduction in these estimates
- Simulating multiple "correlated" random variables
  - Normally distributed
  - Any distributions (copulae)

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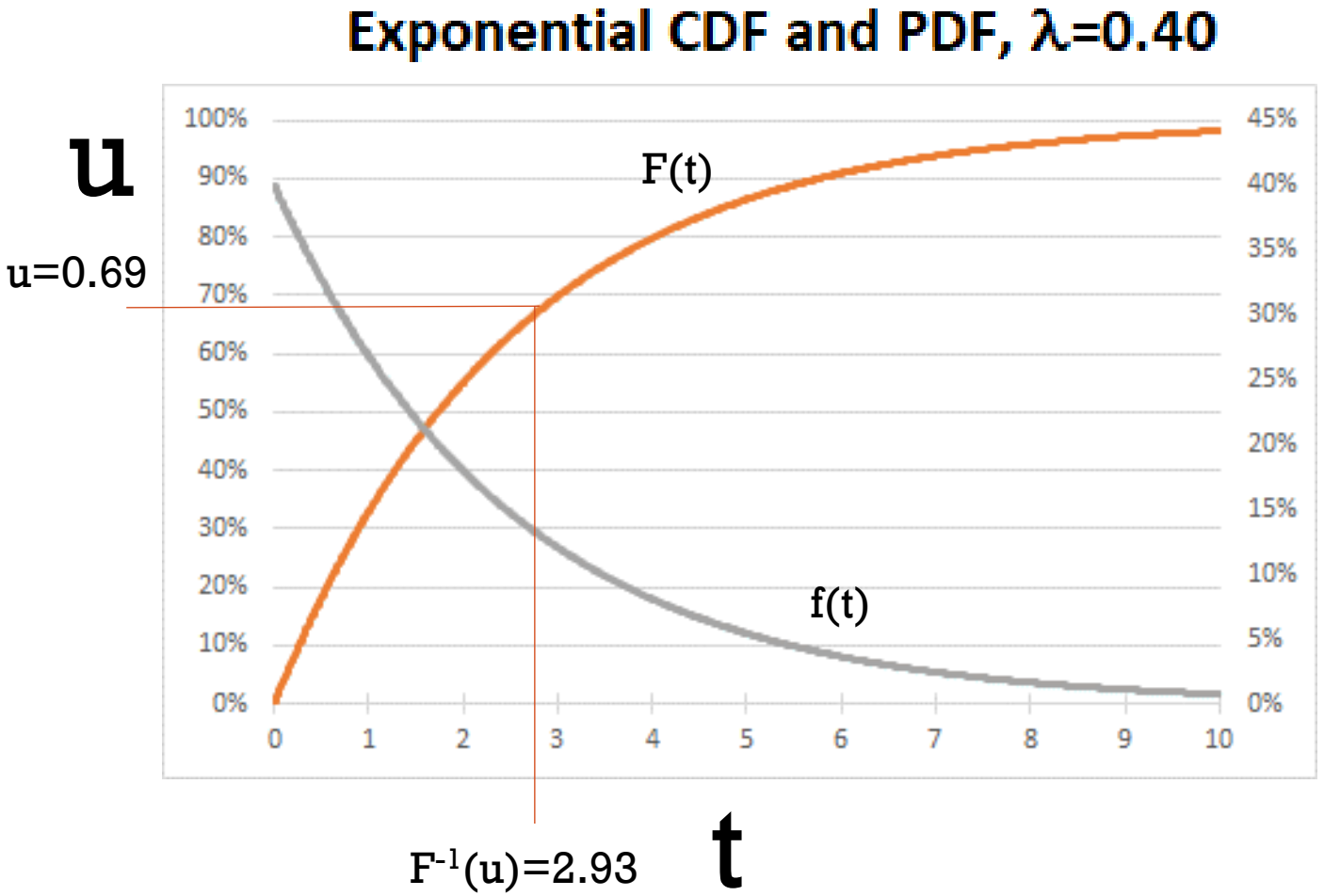
REVIEW QUESTIONS YOU SHOULD KNOW

- In matrix notation, suppose X is a random column vector and its covariance matrix is Σ.
  - Let "1" represent a vector of 1s.
  - Let a,b be vectors.
- Interpret the following:
  - $1'\Sigma 1$
  - $\Sigma 1$
  - $a'\Sigma a$
  - $a'\Sigma b$

  - $s_e^2(X'X)^{-1}$ (a.k.a. design matrix in _____)
  - $(X'X)^{-1}X'Y$ (a.k.a. _____)

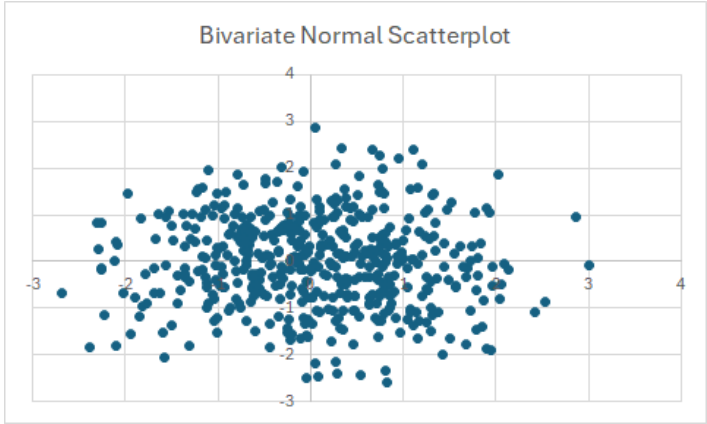
SIMS OF EXPONENTIAL RANDOM NUMBERS

- $f(t) = \lambda e^{-\lambda t}$ with $\lambda=0.4$
- $u = F(t) = 1 - e^{-\lambda t}$
- $t = F^{-1}(u) = -\ln(1 - u)/\lambda$
- Find the mean using simulation

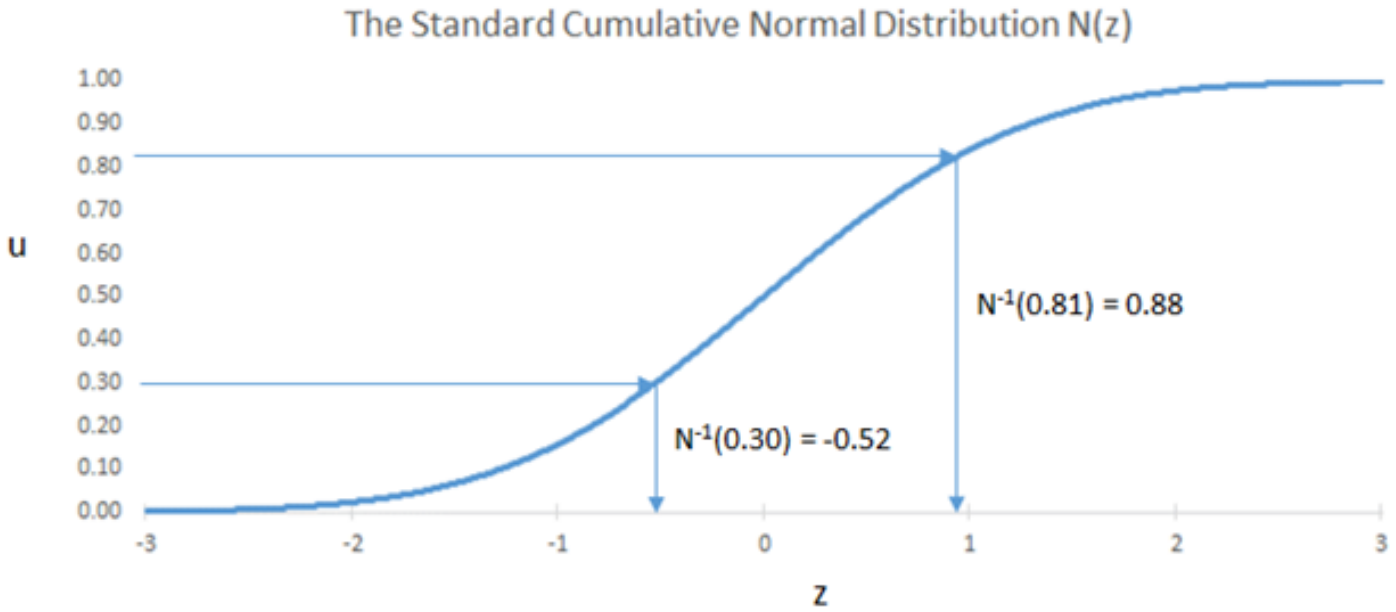


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SIMS OF NORMAL RANDOM NUMBERS



- $z = N^{-1}(u)$
  - N is the standard normal CDF
- z = normsinv(rand( ))
  - Non-analytic
  - High quality approximations available in Excel & Python
- Exercise:
  - Simulate a bivariate normal distribution in Excel



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THE PURPOSE OF SIMULATION

- Application: Consider a function of a single random variable
  - We want to know the probability distribution and/or moments of that function

- The general expression of expectation g(x) of a random variable x can be written
  - $E[g(x)] = \int_{-\infty}^{\infty} g(x)f(x)dx$
  - Where f(x) is the probability density function f(x).

- If f(u) is uniform, i.e. f(u)=1, we have $E[g(u)] = \int_0^1 g(u)du \approx \frac{1}{n}\sum_{i=1}^{n} g(u_i)$
  - Where u_i is a uniformly distributed random number

- Therefore, in the univariate case we can often solve for the expectation as an integral or a numerical integral approximation
  - But in multivariate cases, this becomes impossible, and simulation acts as a substitute for integration

Memorize this!

EVERY SIMULATION DOES IT LIKE THIS



- Generate a large number N of random numbers or vectors z
- Compute outcome values f(z)
- Compute the average value $\bar{X}$ of the f(z) results --- this is your estimate of E[f(z)]
- Compute the standard deviation of the values of f(z). Let's call this "s"
- Compute the *standard error*, which is the standard deviation divided by $\sqrt{N}$, i.e. $s/\sqrt{N}$
- Compute the maximum likely error, which is m times the standard error $= ms/\sqrt{N}$ (m is a matter of choice corresponding to the number of standard errors)
- If the maximum likely error is too large, increase the size of N in proportion to the square of the desired increase in precision.
- The solution is then given by $E[f(z)] = \bar{X} \pm ms/\sqrt{N}$

Let's always use m=2 in this class

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EXAMPLE: SIMULATE AN OIL PAYDAY

- An oil well is expected to produce N barrels of oil next year and sell at a net price of P per barrel.
  - N is normally distributed with mean 1500 and standard deviation 500.
  - P is exponentially distributed with parameter 0.02
- What is the expected net revenue?
  - Ans (1000 sims): 70,779±4,810
  - Ans (1,000,000 sims): 75,036±160

Mean production	1500	E1
Std deviation	500	E2
Price parameter	0.02	E3

What-if Data table

					70,779
1	20,846	<---- *1000 sims*		1	72,889
2	37,474			2	76,658
3	29,993	Average	70,779	3	75,804
4	83,183	StDev	76,059	4	77,355
5	49,719	StErr	2,405	5	75,020
6	136,245	MaxErr	4,810	6	70,605
7	84,026			7	75,371
8	213,901	*1000 sims of average ---->*		8	75,655
9	54,445			9	79,841
10	152,460	Average	75,036	10	75,279
11	101,209	StDev	2,526	11	76,482
12	33,490	StErr	80	12	72,274
13	59,084	MaxErr	160	13	78,914
14	11,233			14	73,803
15	94,806			15	75,418
16	33,359	StDev Ratio		16	75,262
17	84,325		30.11	17	74,684
18	44,009	Squared		18	78,350
19	8,341		907	19	77,779
20	2,603			20	69,593



One Million Sims in 1000 Lines!

=-NORM.INV(RAND(),E1,E2)*LN(RAND())/E3

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SIMULATION IMPROVEMENT TECHNIQUES

- These are techniques designed to increase simulation precision for the same number of sims, or reduce the number of sims required
- Class example: We want to find the value of $E[e^z]$
  - z is the standard normal variate, the true value is _____
  - Start with 1000 simulations.
    - What is your estimate?
    - What should it be?
    - What is the maximum error in the value?
  - The easiest way to improve a simulation is to add more sims
    - The maxerr declines with the square root of the sample size
    - For example, if the sample size doubles, the error variance doubles, but the maxerr is divided by 1.414, the square root of 2.

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MOMENT MATCHING

- Can we improve the model by matching moments?
- Transform your z values so that the mean is 0 and the standard deviation is 1.
  - Subtract the mean and divide each value by the standard deviation
- Does this help?

Moment matching makes the estimate of expectation more accurate (1.66 vs. 1.59), but does not necessarily reduce the computed error.

Expected value of e^z (MOMENT MATCHING)

	z (fixed)	z (match)
Mean	0.0207	0.0000
StDev	0.9340	1.0000
Skew	-0.0219	-0.0219
N	1000	1000

True value		1.65
	Average	MaxErr
Fixed	1.59	0.13
Match	1.66	0.15

	Normals		*Exponentials*	
	z (fixed)	z (match)	z (fixed)	z (match)
1	-0.49	-0.55	0.61	0.58
2	-2.25	-2.43	0.11	0.09
3	0.86	0.89	2.35	2.45
4	-0.68	-0.75	0.51	0.47
5	0.84	0.88	2.32	2.41
6	0.23	0.23	1.26	1.26
7	-0.30	-0.34	0.74	0.71
8	-0.61	-0.68	0.54	0.51
9	-0.69	-0.76	0.50	0.47
10	-1.27	-1.38	0.28	0.25
11	0.09	0.07	1.09	1.07
12	-0.14	-0.17	0.87	0.84
13	0.94	0.98	2.56	2.67
14	0.14	0.12	1.15	1.13
15	1.25	1.31	3.49	3.72



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ANTITHETIC SAMPLING

- Can you improve the model with antithetic sampling?
- Double the number of z-samples by including the negative values in the dataset.

Antithetic sampling improved the max error, but there are twice as many sims. However, the sims are easy!

Also, note that antithetic sampling forces mean and skewness to zero when sampling z-values.

Expected value of e^z (ANTITHETIC)

	z (fixed)	z(anti)	True value		1.65	
Mean	0.0207	0.0000				
StDev	0.9340	0.9340		Average	MaxErr	Adj(N)
Skew	-0.0219	0.0000	Fixed	1.59	0.13	
N	1000	2000	Anti	1.56	0.09	0.13

	Normals		*Exponentials*	
	z (fixed)	z(anti)	z (fixed)	z(anti)
1	-0.49	0.49	0.61	1.63
2	-2.25	2.25	0.11	9.45
3	0.86	-0.86	2.35	0.42
4	-0.68	0.68	0.51	1.97
5	0.84	-0.84	2.32	0.43
6	0.23	-0.23	1.26	0.79
7	-0.30	0.30	0.74	1.35
8	-0.61	0.61	0.54	1.84
9	-0.69	0.69	0.50	2.00
10	-1.27	1.27	0.28	3.57
11	0.09	-0.09	1.09	0.92
12	-0.14	0.14	0.87	1.15
13	0.94	-0.94	2.56	0.39
14	0.14	-0.14	1.15	0.87
15	1.25	-1.25	3.49	0.29



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VARIANCE REDOX: CONTROL VARIATE

- Can we predict E(e^z) better by using z and z^2 as (linear) controls?
  - Let $e^z = b_0 + b_1 z + b_2 z^2 + \varepsilon$
  - Then $E[e^z] = b_0 + b_1 E[z] + b_2 E[z^2]$
    - This is the "mean response"
  - We know *E[z] = 0* and *E[z^2] = 1*
  - Simulate z values and perform the regression
  - Estimated result (1000 sims)
    - 0.84 + 0 + 0.80 = 1.64
    - Correct answer = 1.648721 = $e^{0.5}$

1	z	z2	ez
1.00	-0.42	0.18	0.66
1.00	1.03	1.06	2.80
1.00	-1.83	3.34	0.16
1.00	0.40	0.16	1.49
1.00	0.80	0.63	2.22
1.00	1.38	1.91	3.98
1.00	1.41	1.98	4.09
1.00	-0.53	0.29	0.59
1.00	1.09	1.20	2.99
1.00	-0.59	0.35	0.55

Regression statistics (updates with every sim)

LINEST

	b2	b1	b0
Coeff	0.7966	1.4591	0.8408
Sterr	0.0346	0.0415	0.0513
R2/se	0.96	0.40	
ANOVA	1087.05	97.00	
	340.70	15.20	

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CONTROL VARIATE MAXIMUM ERROR

- Use the "variance of the mean response" to calculate the standard error of the mean estimate in regression
- Define the data matrix X (independent variables) as the pink box below; e^z is the (green) dependent variable
- Recall we are estimating $e^z = b_0 + b_1 z + b_2 z^2 + \varepsilon = X_h' b + \varepsilon$
- Let $b' = (b_0 \quad b_1 \quad b_2)$
  - The estimated regression coefficients, or the "coefficient vector"
- Let $X_h' = (1 \quad 0 \quad 1)$
  - X_h could be any of the rows of the X matrix
  - Or any vector X of interest
  - We are interested in using the known expected values of X_h
- Then the estimated mean of e^z is $\mathrm{E}[e^z] = X_h' b = b_0 + 0 + b_2$.
  - X_h is given but b is measured with error
- The variance of the mean is $Var(\mathrm{E}[e^z]) = s_e^2 X_h'(X'X)^{-1} X_h$
  - s_e^2 is the residual variance from the regression
  - The covariance (matrix) of b is $Cov(b) = s_e^2 (X'X)^{-1}$

Sample data

1	z	z2	ez
1.00	-0.42	0.18	0.66
1.00	1.03	1.06	2.80
1.00	-1.83	3.34	0.16
1.00	0.40	0.16	1.49
1.00	0.80	0.63	2.22
1.00	1.38	1.91	3.98
1.00	1.41	1.98	4.09
1.00	-0.53	0.29	0.59
1.00	1.09	1.20	2.99
1.00	-0.59	0.35	0.55

CALCULATION ILLUSTRATION

Control Variate Method - Estimate E[e^z] using z and z^2 as controls (regressors)

	1000 sims		*Note: "x" is the green box*	
	1	z	z^2	e^z
1	1	0.57	0.33	1.77
2	1	-0.85	0.71	0.43
3	1	0.54	0.29	1.71
4	1	-0.32	0.10	0.73
5	1	0.00	0.00	1.00
6	1	-0.75	0.56	0.47
7	1	1.26	1.58	3.51
8	1	-1.24	1.55	0.29
9	1	-1.65	2.72	0.19
10	1	-0.52	0.27	0.60
11	1	0.45	0.21	1.58
12	1	1.11	1.24	3.05
13	1	0.73	0.54	2.08
14	1	1.06	1.13	2.89
15	1	1.24	1.54	3.45
16	1	-1.12	1.25	0.33
17	1	-1.03	1.07	0.36
18	1	-0.19	0.04	0.83
19	1	1.28	1.65	3.61
20	1	0.28	0.08	1.33

Computation of the Variance of Mean Response

Linest function

0.62384	1.56144	0.98924	0.00000	<------	Coefficients in reverse order
0.01067	0.01517	0.01861	#N/A	<------	StErr Coeff
0.95735	0.47813	#N/A	#N/A	<------	R2/se
7460.04721	997.00000	#N/A	#N/A	<------	ANOVA
5116.33880	227.92482	#N/A	#N/A		
0.98924	1.56144	0.62384		<------	Coefficients in correct order
1.00000	0.00000	1.00000	<-- Xh	1.61308	Estimate of E[Y]

inv(X'X)

0.00151	-0.00005	-0.00050
-0.00005	0.00101	0.00012
-0.00050	0.00012	0.00050

*Covariance of coefficients - inv(X'X) *se2*

0.00035	-0.00001	-0.00012
-0.00001	0.00023	0.00003
-0.00012	0.00003	0.00011

se2*Xh*cov

0.00023	0.00002	0.00000

se2*Xh'*cov*Xh	0.00005
sqrt	0.00725
maxerr	0.01450

Variance of mean response
Standard error of mean response

CONTROL VARIATE REVIEW & DISCUSSION

- The idea of control variates: Find useful regressors to reduce residual error
- The value comes from knowing the expected values of these regressors, so the uncertainty is only in the coefficients
- The covariance of the coefficient vector is easy to compute: $s_e^2(X'X)^{-1}$
  - This provides the standard error of each of the coefficients as well as the covariance
  - The variance of $X_h'b$ (a linear combination of the b's) is therefore $s_e^2 X_h'(X'X)^{-1}X_h$
    - This is the variance of the mean response
- THIS METHOD REDUCES VARIANCE DRAMATICALLY
  - In our example, the maxerr drops to 0.014 for 1000 sims
  - Using moment matching and antithetic sampling, the range was 0.13-0.15 for 1000 sims.

MULTIVARIATE SIMS

- We can always simulate multiple *uncorrelated* variables, the problem is to correlate them
  - *What is the meaning of correlation in this context?*
  - *How do we handle more than a few correlated normal variables?*
  - *How do we handle non-normal variables?*
- Correlation is defined relative to a linear relationship
  - Does not apply to all joint probability distributions
  - Constant parameters (mean, stdev, correl)

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BIVARIATE NORMAL SIMS

- Suppose we have a standard normal variate z_1.
- Can we create z_2 so that its mean is zero, stdev is 1, and correlation with z_1 is ρ?
- Take z_3 independent of z_1
- Let $z_2 = \rho z_1 + z_3\sqrt{1-\rho^2}$
- Test the mean, stdev & correl
- In matrix form, $\underset{\text{Correl}}{(z_1 \quad z_2)} = \underset{\text{Uncorrel}}{(z_1 \quad z_3)} \begin{pmatrix} 1 & \rho \\ 0 & \sqrt{1-\rho^2} \end{pmatrix}$
- **What is the 2x2 matrix called?**

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MULTIVARIATE NORMAL SIMS (5)

- Class project:
  - Create a set of 100 instances of 5 random normal variates that have an *exact* target correlation matrix
  - Target correlation matrix = Common correlation 0.2 for all off-diagonal elements
  - Test your results

	Common correlation				0.2
Target	1.00	0.20	0.20	0.20	0.20
	0.20	1.00	0.20	0.20	0.20
	0.20	0.20	1.00	0.20	0.20
	0.20	0.20	0.20	1.00	0.20
	0.20	0.20	0.20	0.20	1.00
CHOLESKY	1.00	0.00	0.00	0.00	0.00
	0.20	0.98	0.00	0.00	0.00
	0.20	0.16	0.97	0.00	0.00
	0.20	0.16	0.14	0.96	0.00
	0.20	0.16	0.14	0.12	0.95
TEST	1.00	0.20	0.20	0.20	0.20
	0.20	1.00	0.20	0.20	0.20
	0.20	0.20	1.00	0.20	0.20
	0.20	0.20	0.20	1.00	0.20
	0.20	0.20	0.20	0.20	1.00

File: Lecture 4 - Cholesky five.xlsx

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COPULAE

- A multivariate density function is a copula if each of the marginal densities is distributed uniformly on [0,1]
- This allows one to map from one density to another using the CDFs.
- Class exercise:
  - "Correlate" two random uniform variates
  - Use this to correlate two exponential distributions with parameters 0.2 and 0.3
  - Show a scatterplot of your results
    - For zero correlation
    - For 80% correlation

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CONCLUSIONS

- Any random variable can be simulated by the function $F^{-1}(u)$, where F^{-1} is the inverse CDF of u, and u is a uniformly distributed random variable on [0,1]
- To simulate the value of a function of number of uncorrelated random variables (any distributions), follow these steps
  - Generate N values and calculate the mean
  - Calculate the standard deviation, and divide by $\sqrt{N}$ to get the standard error
  - Report the mean as your estimate and your "maximum error" as 2 times the standard error
  - Increase the sample size if more precision is needed
- To correlate normally distributed random z variables, use independent random simulations and multiply the data vector by the transpose of the Cholesky decomposition of the desired covariance matrix
  - (Pearson) Correlation is only defined for normal random variables
- To "correlate" random variables of any distribution, convert correlated normal random numbers to uniform distributions, and then use $F^{-1}(u)$ to produce the desired random variables

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MULTIPLE LINEAR REGRESSION – MATRIX

APPENDIX

- We want to estimate a linear equation of the form
  - $y = X\beta + \varepsilon$
  - Note: Greek letters are used for population values, Roman letters for samples
- Where y and ε are n×1 vector, X is an n×k matrix, and β is a k×1 vector
- The variable y is called the dependent variable, or the response variable, regressand, criterion, predicted variable, measured variable, explained variable, experimental variable, responding variable, outcome variable, or label
- The matrix x represents the set of independent variables, predictors, covariates, regressors, controlled variables, manipulated variables, explanatory variables, exposure variables, risk factors, features, input variables
- Given the data, i.e. X and y, we want to get the equation with best fit by selecting β.
  - Note that if X and u are jointly normally distributed, so is ε
  - We may include a constant term in the regression by allowing the first column of X to be the 1 vector
  - The *best fit* for the sample data would be found by minimizing e′e
- Can we derive the OLS (ordinary least squares) estimate of b?

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DERIVATION

- $\min_b e'e = (y - Xb)'(y - Xb)$
- First simplify
  - Note: you could have used the chain rule or the product rule!
  - $e'e = y'y - y'(Xb) - (Xb)'y + (Xb)'(Xb)$
  - Note these are all scalars, so the transpose does not change the value
  - $e'e = y'y - 2b'X'y + b'(X'X)b$
- Now take the unconstrained derivative with respect to b
  - $\frac{\partial e'e}{\partial b} = 0 = -2X'y + 2X'Xb$
  - $X'Xb = X'y$ *the "normal equations"*
  - $b = (X'X)^{-1}X'y$ *the estimated coefficients*

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SAMPLE REGRESSION OUTPUT (EXCEL)

y	X			e
1	1	1	1	-3.12
1	1	2	4	0.34
2	1	3	9	2.65
3	1	4	16	2.81
5	1	5	25	1.84
8	1	6	36	-0.28
13	1	7	49	-2.55
21	1	8	64	-3.96
34	1	9	81	-2.51
55	1	10	100	4.79

X'X			
	10	55	385
	55	385	3025
	385	3025	25333

inv(X'X)			
	1.3833	-0.5250	0.0417
	-0.5250	0.2413	-0.0208
	0.0417	-0.0208	0.0019

X'y	b	B'Xb	X'e
143	9.7167	143	0.00
1209	-6.6705	1209	0.00
10719	1.0720	10719	0.00

y'y	4,895.00	n	10
b'X'y	4,815.35	k	3
b'X'Xb	4,815.35	e'1	0.00
e'e	79.65		

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9918
RSQ	0.9837
Adj RSQ	0.8362
StErr	3.3733
Observations	10

ANOVA

	df	*SS*	*MS*	*F*	*Signif F*
Regression	3	4,815	1,605	141	0.0000
Residual	7	80	11		
Total	10	4,895			

	Coeff	*StdErr*	*t Stat*	*P-value*
Intercept	9.7167	3.9675	2.4491	0.0442
Linear	-6.6705	1.6570	-4.0256	0.0050
Quadratic	1.0720	0.1468	7.3021	0.0002

Notes

$$Multiple\ R = \sqrt{RSQ}$$

$$RSQ = 1 - e'e(y'y)^{-1}$$

$$Adj\ RSQ = 1 - (1 - RSQ)\left(\frac{n-1}{n-k-1}\right)$$

$$StErr = \sqrt{e'e/(n-k)}$$

$$SS(Regr) = b'X'y$$

$$SS(Resid) = e'e$$

$$MS(Regr) = \frac{SS(Regr)}{k}$$

$$MS(Resid) = \frac{SS(Resid)}{n-k}$$

$$F = \frac{MS(Regr)}{MS(Resid)} \quad d.f. = \{k, n-k\}$$

$$Var(b) = \frac{e'e}{n-k}(X'X)^{-1}$$

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LINEST OUTPUT FIELDS (EXCEL)

- If you set the third argument (const) and fourth argument (stats) to TRUE, LINEST will return five rows:
- **Row 1**: Slopes and intercept (in reverse order).
- **Row 2**: Standard errors of the coefficients (slope(s) and intercept).
  → Smaller values indicate more reliable estimates.
- **Row 3**:
  - **R^2 (coefficient of determination)**: Explains how much of the variation in Y is explained by X. Closer to 1 = better fit.
  - **Standard error of Y estimate (SEy)**: Standard deviation of the residuals; how far observed values typically fall from the regression line.
- **Row 4**:
  - **F statistic**: Measures overall significance of the regression (larger F = stronger evidence the model is meaningful).
  - **Degrees of freedom (df)**: Used in significance testing, based on sample size and number of predictors.
- **Row 5**: (Analysis of variance or ANOVA)
  - **Regression sum of squares (SSR)**: Variation explained by the model.
  - **Residual sum of squares (SSE)**: Variation not explained by the model.